

1. Why would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system?

- Advocating for transparency or explainability in an ML/AI system is important because it helps reduce the “black-box” feeling that many modern models create. When decisions affect real people, such as in healthcare, hiring, or law, it becomes essential to understand how those decisions were made. Explainability makes AI models not only accurate but also understandable and trustworthy.
- For users, explainability builds confidence because they can see the reasoning behind a system’s predictions. It helps them spot potential bias, understand why a decision was made, and feel assured that the outcome is fair and not random. This increases trust and makes users more comfortable relying on AI systems, especially in sensitive situations.
- For organizations, transparency supports fairness, accountability, and responsible development. It allows teams to identify and fix biased data, improve model behavior, and ensure that decisions are driven by meaningful features. This strengthens the organization’s credibility, aligns with ethical and legal expectations, and makes it easier to deploy and maintain AI systems responsibly.

2. What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like.

A fair system ensures that all individuals are treated equitably, without discrimination based on sensitive attributes such as gender, race, or age. Fairness in ML/AI often involves detecting and correcting biases in the data or model so that outcomes do not unfairly disadvantage any group. For example, in a fraud detection system, fairness means that users with similar financial behavior have the same likelihood of being flagged, regardless of their age or the community they belong to. Achieving fairness would involve ensuring the model does not rely on sensitive attributes, testing the system across different groups, and adjusting it so all users receive consistent and unbiased outcomes.

Another Example is Unbiased facial analysis systems. A facial recognition system could perform better on certain skin tones or genders if the training data is imbalanced. Achieving fairness requires using diverse datasets, monitoring model accuracy across subgroups, and adjusting algorithms to ensure equal performance for all. In both cases, fairness is about designing the system so that decisions are consistent, equitable, and do not discriminate against protected groups. Techniques include removing biased features, applying fairness constraints, and continuously monitoring outcomes for different groups.

3. How do you see concepts like bias, fairness, accountability, and transparency-related?

Bias happens when an AI system gives results that unfairly disadvantage certain groups, often because the training data is not balanced. Fairness makes sure that outcomes are equal for all users, no matter their race, gender, or age. Transparency shows how the system makes decisions so that any bias can be spotted and fixed, and accountability makes organizations responsible for the effects of their AI systems.

All these ideas together form the basis of responsible AI. Transparency allows people to check the system, fairness ensures fair results, bias points out problems to fix, and accountability builds trust. Together, they make AI systems that work well and are fair and trustworthy for

everyone.

4. Give an example of a system that is biased but fair.

A personalized health recommendation system can be biased but still fair. The system may favor users who consistently track their health metrics, giving an advantage to individuals who are more tech-savvy or actively monitor their health. This creates bias because not everyone has the same habits or access to tracking tools.

However, the system remains fair because the same rules and criteria are applied to all users. The recommendations are designed so that the overall benefits or utility are equitable, ensuring that anyone who provides similar health data receives the same level of guidance and value from the system.